

Table 2: Statistics and initial feature representations of the benchmark datasets.

Dataset	Modality	Samples	Classes	Feature Dim.	Initial Feature
Synthetic	Tabular	1,000	10	3	3D Gaussian features
20 Newsgroups	Text	7,532	20	384	Sentence embedding
MNIST	Image	10,000	10	784	Pixel features
Fashion-MNIST	Image	10,000	10	784	Pixel features

A. EXPERIMENTAL DETAILS

This appendix section provides detailed experimental settings for datasets, feature preprocessing, baseline configurations, PISC, and evaluation. Unless otherwise specified, all methods follow the common-feature protocol described below, and all experiments use the same data splits and a fixed random seed of 42. Experiments are conducted on CPUs with `OMP_NUM_THREADS = 1`.

A.1. Datasets and Feature Construction

We evaluate PISC on four datasets covering tabular, textual, and image modalities. Their statistics and initial feature representations are summarized in Table 2. The datasets are described as follows:

- **Synthetic.** We generate a three-dimensional synthetic dataset consisting of 10 Gaussian clusters, with 100 samples per cluster and a standard deviation of 0.5. The resulting 1,000 samples are directly used as the initial feature representations.
- **20 Newsgroups.** We use 7,532 documents from the 20 Newsgroups dataset [18], covering 20 semantic categories. Each document is encoded using the pretrained `all-MiniLM-L6-v2` sentence encoder, producing a 384-dimensional representation.
- **MNIST.** We use 10,000 handwritten digit images from 10 classes [19]. Each 28×28 grayscale image is flattened into a 784-dimensional pixel vector.
- **Fashion-MNIST.** We use 10,000 grayscale images from 10 apparel categories [20]. Each 28×28 image is flattened into a 784-dimensional pixel vector.

For each real-world dataset, we use a fixed validation/test split shared by all methods. The validation split is used exclusively for hyperparameter selection, while the test split is reserved for final evaluation.

A.2. Common Feature Protocol

To isolate the effect of the clustering procedures, PISC and the applicable baseline methods operate on a common feature space. The preprocessing pipeline is:

1. ℓ_2 -normalize the initial feature representation;
2. project the normalized features into a three-dimensional UMAP embedding [17];
3. reuse the resulting UMAP-3D representation for all methods that support direct feature-vector input.

UMAP is configured with `n_components = 3`, `n_neighbors = 15`, `min_dist = 0.1`, and `random_state = 42`. The UMAP embedding is computed once for each dataset and then shared across PISC and all applicable baselines. Thus, these methods receive identical input representations and preprocessing. The target number of clusters is fixed to the ground-truth number of classes for all methods. Hyperparameters requiring selection are determined using the validation split and fixed before test-set evaluation.

Exception: PRO-DSC on MNIST. PRO-DSC [26] on MNIST follows its official evaluation configuration and therefore uses the original image features scaled to $[0, 1]$ rather than the shared UMAP-3D representation. This is the only exception to the common-feature protocol.

A.3. Baseline Configurations

We compare PISC with six classical clustering methods and four deep clustering methods. The detailed configurations are provided below. Unless explicitly specified, parameters not listed follow the default settings of the corresponding implementations.

A.3.1. Classical Clustering Baselines

All classical baselines are implemented using `scikit-learn` [29] and operate on the shared UMAP-3D representations.

- **K-means** [21]: `n_clusters = K`, `n_init = 10`, and `random_state = 42`.
- **Agglomerative Clustering** [22]: Ward linkage with `n_clusters = K`.
- **Spectral Clustering** [13]: nearest-neighbor affinity with `n_neighbors = 15`, `assign_labels = "kmeans"`, and `random_state = 42`.

- **Gaussian Mixture Model (GMM)** [24]: `n_components = K`, `covariance_type = "diag"`, `max_iter = 200`, and `random_state = 42`.
- **BIRCH** [25]: `threshold = 0.5`, with the final number of clusters set to K .
- **DBSCAN** [23]: `eps = 0.6` and `min_samples = 10`. Samples assigned the noise label -1 are retained as an additional pseudo-cluster during evaluation.

A.3.2. Deep Clustering Baselines

Whenever supported by the corresponding implementation, all deep clustering baselines receive the shared UMAP-3D representations. Their configurations are as follows:

- **PRO-DSC** [26]: trained for 12 epochs with a batch size of 512. The learning rates for the feature and clustering networks are both 10^{-4} . We use SGD with momentum 0.9, hidden dimension 256, latent dimension 128, `pieta = 0.175`, `eps = 0.1`, `wd1 = 10^{-4}`, and `wd2 = 5 \times 10^{-3}`. On MNIST, the original $[0, 1]$ pixel features are used following the official evaluation configuration.
- **DEC+BRB** [27]: follows the default BRB configuration, using Adam with a batch size of 256. The model is pretrained for 250 epochs with a learning rate of 10^{-4} , followed by 200 clustering epochs with a learning rate of 10^{-3} .
- **IDEA+BRB** [27]: follows the default BRB configuration, using Adam with a batch size of 256. The model is pretrained for 250 epochs with a learning rate of 10^{-4} , followed by 200 clustering epochs with a learning rate of 10^{-3} .
- **DCBoost** [28]: trained for 50 epochs using Adam with a batch size of 256 and a learning rate of 10^{-3} .

The random seed is fixed to 42 for all deep clustering baselines.

A.4. PISC Configuration

PISC consists of five main stages: pseudo-prototype generation, prototype grouping and representative selection, prototype-induced response construction, response sparsification, and final clustering. Unless otherwise specified, the same pipeline is applied to all datasets.

A.4.1. Pseudo-prototype Generation

Starting from the shared UMAP-3D representations, we generate an overcomplete set of 200 pseudo-prototypes using K-means, i.e., $|\mathcal{P}| = 200$, with `n_init = 10` and `random_state = 42`.

A.4.2. Prototype Grouping and Representative Selection

The number of retained representatives is controlled by the prototype grouping threshold τ . Different values of τ produce different prototype groupings and, consequently, different representative-set sizes $|\mathcal{R}|$. We construct 120 candidate threshold values uniformly spaced between 0.01 and 0.60 using `linspace(0.01, 0.60, 120)`. For each candidate τ , prototype grouping is performed using the greedy clique-cover procedure with the random seed fixed to 42. Within each resulting group, at most two representatives are retained, i.e., `MAX_REPRESENTATIVES_PER_GROUP = 2`. The representative selection rule is:

- if a group contains one or two pseudo-prototypes, all members are retained;
- if a group contains more than two pseudo-prototypes, the pair with the minimum pairwise distance is retained.

Thus, the final representative-set size is determined jointly by τ and the induced group structure.

A.4.3. Prototype-induced Representation

After representative selection, each sample is mapped to a prototype-response representation. For the negative-Euclidean setting, the response between sample $\mathbf{x}$ and representative prototype $\mathbf{p}$ is defined as $r(\mathbf{x}, \mathbf{p}) = \exp(-\|\mathbf{x} - \mathbf{p}\|_2)$. For the cosine-similarity setting, the response is defined as $r(\mathbf{x}, \mathbf{p}) = \exp(\cos(\mathbf{x}, \mathbf{p}))$.

We then apply row-wise top- κ sparsification. Specifically, only the κ strongest prototype responses for each sample are retained, while all remaining entries are set to zero. The resulting sparse prototype-induced representations are finally clustered using K-means with `n_clusters = K`, `n_init = 10`, and `random_state = 42`.

A.4.4. Hyperparameter Selection

The sparsity parameter is selected from $\kappa \in \{5, 8, 10, 15, 20, 30\}$. The threshold τ is selected to obtain a representative-set size as close as possible to a predefined target. Both τ and κ are selected exclusively on the validation split and fixed before test-set evaluation. The configurations used in the main experiments are summarized in Table 3.

Because the representative-set size is induced by the prototype-group structure, the actual number of retained representatives can differ slightly from the target. The resulting representative-set sizes in the main experiments are 160, 99, 153, and 138 for Synthetic, 20 News-groups, MNIST, and Fashion-MNIST, respectively. In addition, PISC does not require task-specific neural-network training after the initial feature extraction. The remaining computation consists entirely of low-dimensional prototype construction, prototype grouping, response computation, sparsification, and final clustering.

Table 3: PISC configurations used in the main experiments.

Dataset	Target $ \mathcal{R} $	κ	Similarity
Synthetic	160	5	Negative Euclidean
20 Newsgroups	100	8	Negative Euclidean
MNIST	160	10	Negative Euclidean
Fashion-MNIST	140	20	Cosine

Table 4: Clustering performance (%) on three real-world datasets using the original high-dimensional representations. The final row reports the performance of PISC using the UMAP-3D representations adopted in the main experiments. The best result for each metric is highlighted in bold.

Method	20 Newsgroups			MNIST			Fashion-MNIST		
	F1	ARI	NMI	F1	ARI	NMI	F1	ARI	NMI
DBSCAN	0.14	0.00	0.14	0.00	0.00	0.00	1.73	1.89	10.21
BIRCH	47.22	30.33	48.91	67.14	60.63	71.14	21.20	12.61	42.92
K-means	51.13	35.01	51.47	53.65	37.91	50.17	51.24	40.91	60.47
Agglomerative	46.04	33.02	49.44	67.14	60.63	71.14	52.59	41.70	62.13
GMM	50.51	34.06	50.95	40.23	20.07	30.92	47.79	37.81	56.45
Spectral	39.89	27.51	54.05	65.20	54.56	70.02	53.09	41.61	62.89
PRO-DSC	37.05	18.45	29.94	57.60	41.68	53.12	55.59	42.29	59.22
DEC+BRB	5.05	7.79	21.97	30.66	15.99	29.32	11.36	14.54	28.64
IDEC+BRB	5.17	7.84	22.13	11.47	12.99	21.66	45.42	34.09	50.52
DCBoost	26.39	10.26	24.57	52.67	35.47	48.73	38.33	28.45	47.86
PISC (Original Features)	46.63	29.42	46.24	43.67	27.53	41.14	49.42	35.99	52.74
PISC (UMAP-3D)	55.80	39.97	55.43	94.09	87.81	87.89	61.54	49.74	62.77

A.5. Evaluation Protocol

We evaluate all methods using F1 score, Adjusted Rand Index (ARI), and Normalized Mutual Information (NMI). Because predicted cluster indices are permutation invariant with respect to ground-truth class labels, we first use the Hungarian algorithm to align predicted clusters with the reference classes before computing F1. ARI and NMI are computed directly from the predicted and ground-truth assignments because both metrics are invariant to cluster-label permutations. All metrics are reported as percentages. No test-set information is used for hyperparameter selection or model configuration. For reproducibility, we follow the following evaluation protocol:

- hyperparameters are selected exclusively using the validation split;
- the selected configurations are fixed before evaluating the test split;
- the same data split is used across all applicable methods;
- the random seed is fixed to 42;
- `OMP_NUM_THREADS = 1` is used throughout the experiments;
- all experiments are conducted on CPUs.

B. ADDITIONAL EXPERIMENTS

B.1. Effect of Input Representation

Table 4 further examines the influence of the input representation on clustering performance. Across all three real-world datasets, PISC consistently performs better with the UMAP-3D representations than with the original high-dimensional features. The improvement is observed across F1, ARI, and NMI, indicating that the benefit of manifold projection is not specific to a single evaluation metric. Among the three datasets, the effect is most pronounced on MNIST, while 20 Newsgroups exhibits a more moderate but still consistent improvement. Fashion-MNIST shows an intermediate pattern, suggesting that the extent to which PISC benefits from dimensionality reduction depends on the structure of the original feature space.

The baseline results on the original representations provide additional context for this comparison. Several classical clustering methods remain reasonably competitive in the original feature space, whereas others, particularly density-based and some deep clustering approaches, exhibit substantial degradation. This variation indicates that high-dimensional representations affect different clustering mechanisms differently. In particular, methods that rely strongly on local distance relationships or learned latent partitions can be sensitive to the geometry and sparsity of the original feature space.

Table 5: Interaction between UMAP-based dimensionality reduction and top- κ sparsification on 20 Newsgroups and MNIST. All values denote F1 scores (%). Δ_{Sparse} measures the gain from applying top- κ sparsification under the same feature representation, whereas Δ_{UMAP} measures the gain from replacing the original features with UMAP-3D representations under the same sparsification setting.

Feature representation	20 Newsgroups			MNIST		
	w/o Top- κ	w/ Top- κ	Δ_{Sparse}	w/o Top- κ	w/ Top- κ	Δ_{Sparse}
Original features	43.79	45.73	+1.94	46.11	67.18	+21.07
UMAP-3D	48.01	55.80	+7.79	73.51	94.09	+20.58
Δ_{UMAP}	+4.22	+10.07	–	+27.40	+26.91	–

Table 6: Ablation study of representative prototype selection strategies on 20 Newsgroups and MNIST. “#Prototypes” denotes the number of retained representative prototypes. All performance values are reported in percentages. The best result in each column is highlighted in bold.

Strategy	20 Newsgroups				MNIST			
	#Prototypes	F1	ARI	NMI	#Prototypes	F1	ARI	NMI
Retain all prototypes	200	55.65	32.68	55.02	200	87.41	78.33	83.09
Random pair	99	47.45	38.04	53.04	153	89.77	82.63	85.17
Farthest pair	99	48.81	37.88	52.90	153	91.57	84.71	86.54
Closest pair (Ours)	99	55.80	39.97	55.43	153	94.09	87.81	87.89

For PISC, we compare the original representations with UMAP-3D while keeping the remaining clustering procedure unchanged. The consistent improvement under UMAP-3D suggests that the manifold representation provides a more favorable space for subsequent pseudo-prototype generation, prototype grouping, and sparse response construction. By concentrating the data structure into a compact representation, UMAP can make local relationships between samples and prototypes easier to capture, which in turn benefits representative selection and top- κ sparsification.

Overall, these results indicate that the input representation plays an important role in the effectiveness of PISC. UMAP-3D does not merely reduce dimensionality, but empirically provides a representation space that is better aligned with the prototype-induced clustering procedure. This observation is consistent with the ablation results reported in the main text and further supports the use of manifold representation as an integral component of PISC.

B.2. Ablation Study

B.2.1. Ablation on Representation and Sparsification

Table 5 examines the individual and joint effects of UMAP-based manifold representation and top- κ sparsification. On 20 Newsgroups, both components improve clustering performance when applied individually, while their combination provides a noticeably stronger improvement. This positive interaction suggests that sparsification becomes more effective after the data are mapped to a lower-dimensional manifold, where retaining the strongest prototype responses can better emphasize discriminative similarity relationships.

MNIST exhibits a different interaction pattern. Both UMAP-based representation and top- κ sparsification provide substantial gains individually, and their combination further improves the clustering performance. In contrast to 20 Newsgroups, the marginal contribution of either component changes only slightly when the other is introduced, suggesting that the two components provide largely complementary benefits on MNIST. Overall, the ablation confirms that both manifold construction and sparse prototype responses are important to the effectiveness of PISC, while their interaction depends on the characteristics of the underlying data.

B.2.2. Ablation on Prototype Selection

Table 6 compares four representative prototype selection strategies: retaining all pseudo-prototypes, randomly selecting two prototypes from each group, selecting the farthest pair, and selecting the closest pair used in PISC. The closest-pair strategy consistently achieves the strongest performance across the evaluated metrics and datasets. At the same time, retaining all pseudo-prototypes does not consistently improve clustering quality, indicating that a larger prototype set is not necessarily more informative. These results demonstrate that representative prototype selection is an important component of the prototype reduction process.

The comparison among the three pair-selection strategies further isolates the effect of the selection criterion because they retain the same number of representatives within each dataset. The closest-pair strategy consistently outperforms both random-pair selection and farthest-pair selection, suggesting that the identity of the retained prototypes is critical to the quality of the induced similarity representation. Within each group, selecting mutually similar prototypes favors locally consistent representatives rather than maximizing prototype separation. The results therefore support the use of the closest-pair criterion as a means of reducing prototype redundancy while preserving clustering-relevant structure.